

Task 3: Heuristic Graph Pathfinding Agent

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Summary

This benchmark compares Dijkstra, A* with the Manhattan heuristic, and the bonus tabular Q-learning agent on four reproducible 4-connected grid mazes.

All configurations are generated with fixed seeds and are verified reachable by the generator before search.

Implemented Components

- `environment/grid.py`: bounded grid, obstacle handling, 4-connected movement, and admissible distance heuristics.
- `environment/generators.py`: Easy, Medium, Hard, and Cul-de-Sac benchmark mazes.
- `algorithms/dijkstra.py`: optimal uniform-cost baseline.
- `algorithms/astar.py`: optimal heuristic-guided search with selectable heuristics.
- `algorithms/q\_learning.py`: bonus tabular Q-learning training and greedy path extraction.
- `visualization/plotter.py`: per-maze comparison panels and aggregate metric charts.

Metrics

`Steps` is the final path length, `Runtime (ms)` includes Q-learning training and inference for the RL row, and `Expanded` is the algorithm-specific explored-node count.

Configuration	Algorithm	Solved	Steps	Runtime (ms)	Expanded	Path cost
Easy (15×15, 10% density)	Dijkstra	Yes	28	1.6694	200	28.0
Easy (15×15, 10% density)	A* (Manhattan)	Yes	28	0.9907	169	28.0
Easy (15×15, 10% density)	Q-Learning (RL Agent)	Yes	28	807.9473	29	28.0
Medium (20×20, 25% density)	Dijkstra	Yes	52	2.9313	296	52.0
Medium (20×20, 25% density)	A* (Manhattan)	Yes	52	2.7465	269	52.0
Medium (20×20, 25% density)	Q-Learning (RL Agent)	Yes	52	1923.8360	53	52.0
Hard (25×25, 35% density)	Dijkstra	Yes	130	7.1828	445	130.0
Hard (25×25, 35% density)	A* (Manhattan)	Yes	130	6.7323	395	130.0
Hard (25×25, 35% density)	Q-Learning (RL Agent)	Yes	130	5532.4533	131	130.0
Cul-de-Sac Trap (20×20, U-shaped)	Dijkstra	Yes	38	5.6548	353	38.0
Cul-de-Sac Trap (20×20, U-shaped)	A* (Manhattan)	Yes	38	5.4206	295	38.0
Cul-de-Sac Trap (20×20, U-shaped)	Q-Learning (RL Agent)	Yes	38	1453.3438	39	38.0

Findings

- Dijkstra and A* produce equal optimal path costs on every solvable benchmark maze.

- A* expands no more nodes than Dijkstra in the tested configurations, while using the goal-directed Manhattan heuristic.
- Q-learning is stochastic model-free training; its reported runtime includes training and greedy extraction, while its expanded count measures inference states only.

Exported Visualizations

- `artifacts/cul\_de\_sac\_trap\_comparison.png`
- `artifacts/easy\_comparison.png`
- `artifacts/hard\_comparison.png`
- `artifacts/medium\_comparison.png`
- `artifacts/metrics\_comparison.png`

Validation

The repository test suite contains 24 unit and integration tests covering grid behavior, heuristic validity, path correctness, optimality, unsolvable grids, and all four benchmark configurations.

Reproduction

```
```powershell
.\.venv\Scripts\python.exe benchmark.py
```
```